



**B.Tech. Degree II Semester Regular and Supplementary /
I Semester Supplementary Examination May 2017**

**CE/EE/ME/SE AS 15-1202 A & CS/EC/IT AS 15-1102 B
ENGINEERING CHEMISTRY
(2015 Scheme)**

Time: 3 hours.

Maximum Marks: 60

**PART A
(Answer ALL questions)**

(10 × 2 = 20)

- I. (a) What are Schottky defects? Illustrate with one example.
 (b) Explain the principle of ESCA.
 (c) Explain how FT-IR spectra is useful for molecular structure determination.
 (d) What is a salt bridge? Give one application scenario.
 (e) Explain the principle of buffer action.
 (f) Assuming an activation energy of 100 kJ/mole for a reaction under reflux at 1 atm pressure, estimate the change in the rate constant for the reaction when the medium of reaction changed from ethanol (b.p=78.2 °C) to water (b.p=100 °C).
 (g) What is eutectic composition? Give one example.
 (h) What are electrolytic concentration cells? Calculate the emf of the following cell. $\text{Cu(s)}|\text{Cu}^{2+}(0.2\text{M})||\text{Cu}^{2+}(2.0\text{M})|\text{Cu(s)}$.
 (i) What are excess functions and regular solutions?
 (j) Explain the chemistry of vulcanization.

PART B

(4 × 10 = 40)

- II. (a) State and explain Beer-Lambert law. Explain its limitations. (5)
 (b) How can you differentiate intramolecular and intermolecular hydrogen bonding by FT-IR spectroscopy? (3)
 (c) O_2 or N_2 are microwave inactive whereas CO gives a microwave spectrum. Explain why. (2)

OR

- III. (a) Discuss about a method for the determination of surface area of porous solid samples. (3)
 (b) Explain spin-spin splitting using the ^1H NMR spectrum of EtOH. (3)
 (c) Explain the molecular field theory of magnetic materials. (4)
- IV. (a) Explain the Bronsted-Lowry and Lewis concepts of acids and bases. (4)
 (b) A sample of aqueous solution of a household cleaning formulation was analyzed for OH^- ions and the concentration of OH^- was found to be 0.0025 M. What will be the concentration of H^+ ions? State whether the solution is acidic or basic in nature. (2)
 (c) Calculate the pH of a 1000 ml solution of 0.25 M acetic acid and 0.25 M sodium acetate per litre. What would be the change in pH when 1 ml of 1 M HCl was added to this solution? (The dissociation constant of acetic acid is 1.8×10^{-5} at 25 °C.) (4)

OR

(P.T.O.)

- V. (a) Distinguish between chemical and electrochemical corrosion. (3)
 (b) What is bio-corrosion? Explain the mechanism and the cell reaction. (2)
 (c) Give a brief account of methods of corrosion prevention. (5)
- VI. (a) Distinguish between (i) open, closed and isolated systems; (ii) isothermal and adiabatic changes. (4)
 (b) Explain Trouton's rule. Predict the standard molar enthalpy of vaporization of bromine. (Bromine boils at 59.2 °C). (3)
 (c) Haber-Bosch process is considered as the detonator of the population explosion by aiding agriculture revolution and food production. It is estimated that 80% of the nitrogen found in human tissues originate from this process. What is the reaction Gibbs energy when the partial pressure of the N_2 , H_2 and NH_3 (treated as ideal gases) are 3 bar, 1 bar and 4 bar respectively? What is the spontaneous direction of the reaction in this case? (Standard Gibbs energy of formation of ammonia as -1616.5 kJ/mol at 298 K.) (3)
- OR**
- VII. (a) Sketch the phase diagram of water and explain. (3)
 (b) On the phase diagram mark the lowest and highest temperature which liquid water exist. (2)
 (c) Contrary to what is generally observed melting temperature of water ice falls as the pressure is applied to about 2 bar. Explain why. (2)
 (d) In order to sustain aquatic life an oxygen concentration of 4 mg/L is required in the water. Calculate the minimum partial pressure of oxygen in the atmosphere to achieve the above oxygen concentration in water. (Henry's law constants (K/Torr) for gases at 25 °C in water: Methane -3.14×10^5 ; $CO_2 - 1.25 \times 10^6$; $H_2 - 5.34 \times 10^7$; $N_2 - 6.51 \times 10^7$ and $O_2 - 3.30 \times 10^7$). (3)
- VIII. (a) Explain the mechanism of anionic and cationic polymerization. (4)
 (b) What are stereoregular polymers? Discuss their methods of preparation. (3)
 (c) Discuss about ways of plastic waste disposal. (3)
- OR**
- IX. (a) What are lubricants? Discuss about their classification and applications. (4)
 (b) Explain the theory of setting and hardening of cement. (3)
 (c) Discuss about the properties of refractories. (3)